Applied Unsupervised Learning in Python

Today's goal • 3/3 items

Module 1
Basic Unsupervised Learning Methods

Module 2
Clustering

Module 3
Unsupervised Methods for Text Analysis

Lesson 1: Text Representations

How to Represent Text as a Vector: A Typical Workflow
Video • 21 min

Time to Practice: Representing Text as a Vector
Graded Assignment • Grade: 100%

Text Processing in SciKit-Learn
Video • 13 min

Time to Practice: Text Processing in SciKit-Learn
Graded Assignment • Grade: 100%

Lesson 2: Topic Modeling Methods & Techniques

Introduction to Topic Modeling
Video • 18 min

Latent Dirichlet Allocation (LDA)
Video • 7 min

Using LDA with Scikit-Learn
Video • 11 min

Time to Practice: Latent Dirichlet Allocation (LDA)
Graded Assignment • Grade: 100%

Non-Negative Matrix Factorization (NMF)
Video • 19 min

Time to Practice: Non-Negative Matrix Factorization (NMF)
Graded Assignment • 15 min

Lesson 3: Word Embeddings Techniques

Word Embeddings Technique #1: Word2vec
Video • 21 min

Word Embeddings Technique #2: Glove
Video • 16 min

Time to Practice: Word2vec & Glove
Graded Assignment • 15 min

Module 3 Wrap-Up

Introduction to Module 3 Assignment: Text Representations, Topic Modeling, and Word Embeddings
Reading • 5 min

Create & Submit Module 3 Assignment
Programming Assignment • Grade: 100%

Assignment 3 Followup
Dialogue • 15 min

Module 3 Optional Readings & Resources
Reading • 5 min

Module 4
Applications and Variants of Unsupervised Learning

Lesson 1: Unsupervised and Supervised Learning

Applying Unsupervised Learning to Supervised Learning Tasks
Video • 19 min

Time to Practice: Applying Unsupervised Learning to Supervised Learning Tasks
Graded Assignment • 15 min

Lesson 2: Handling Missing Data

Imputation of Missing Data
Video • 14 min

Imputation with Scikit-Learn
Video • 19 min

Time to Practice: Imputation of Missing Data
Graded Assignment • 15 min

Lesson 3: Semi-Supervised Learning

A Brief Introduction to Semi-Supervised Learning
Video • 19 min

Label propagation with scikit-learn
Video • 15 min

Time to Practice: Semi-Supervised Learning
Graded Assignment • 15 min

Lesson 4: Self-Supervised Learning

A Brief Introduction to Self-Supervised Learning
Video • 17 min

Time to Practice: Self-Supervised Learning
Graded Assignment • 15 min

Lesson 5: Module 4 Wrap-Up

Introduction to Module 4 Assignment: Applying Methods and Techniques for Data Imputation and Semi-Supervised Learning
Reading • 5 min

Create & Submit Module 4 Assignment
Programming Assignment • Grade: 100%

Assignment 4 Followup
Dialogue • 15 min

Module 4 Optional Readings & Resources
Reading • 5 min

Course Conclusion
Video • 3 min

Post-Course Survey
Reading • 10 min

Introduction to Module 3 Assignment: Text Representations, Topic Modeling, and Word Embeddings

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For the Module 3 programming assignment, you will focus on topic modeling using a dataset of recipes. You will be doing a number of tasks pertaining to topic modeling, such as obtaining top topic terms with Non-negative Matrix Factorization (NMF) and using Latent Semantic Indexing to compare recipe ingredient lists.

In modern-day applications, topic modeling can be very useful when attempting to understand themes/subjects across a large number of documents.

1. **First**, you are going to compute topic models using non-negative matrix factorization. You'll write a function to return the top cluster terms for each of ten topics found by NMF on the recipe dataset.
2. **Second,** you are going to use Latent Semantic Indexing (LSI) to use a model trained from the recipes dataset to find the ingredient lists most and least similar to a given single ingredient list.
3. **Third,** you are going to explore how to estimate semantic coherence of a topic, using a provided word embedding to compute the semantic similarity of a topic's terms. In the first task, you created a list of top topic terms for the first 10 topics in the recipe dataset derived using NMF. Now, you are going to use a pre-trained word embedding to estimate the semantic coherence of those topics from task 1.
4. **Fourth,** you will explore the fascinating phenomenon of 'semantic arithmetic' that has been observed in common word embeddings, using the provided word embedding trained from Amazon review data.

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